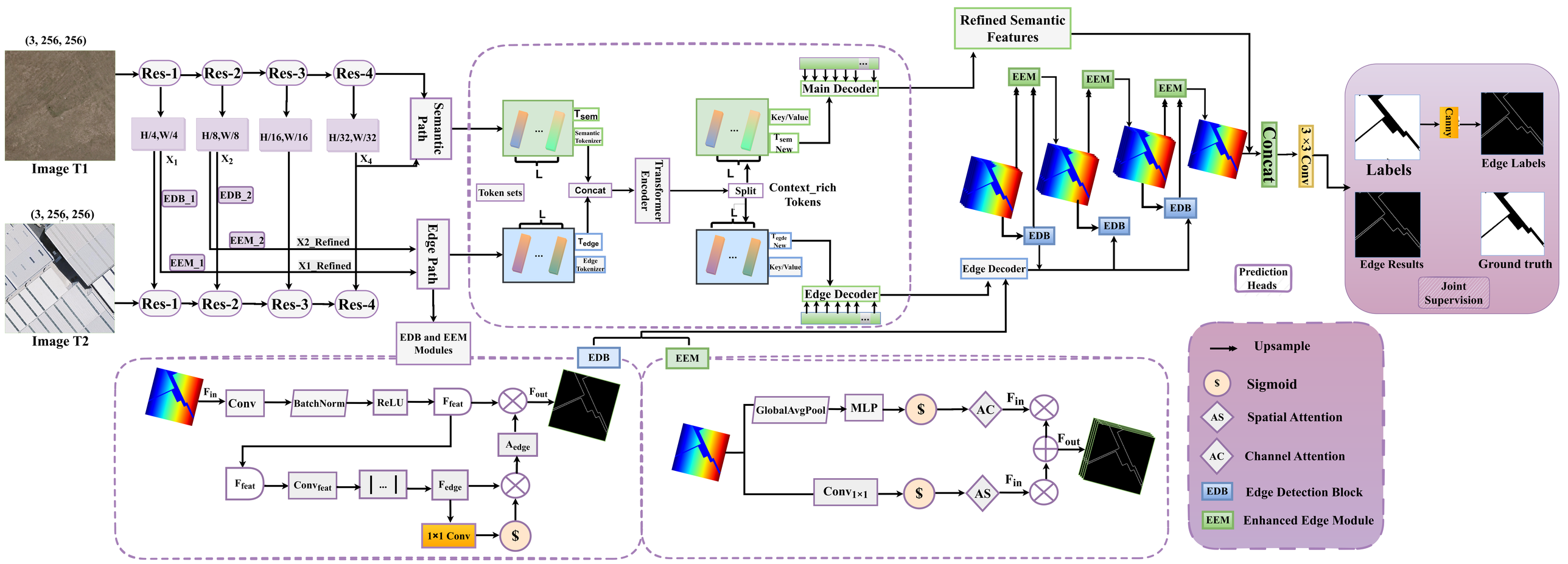
(3, 256, 256)
Image T1
(3, 256, 256)
Image T2
Res-1
Res-2
Res-3
Res-4
H/4,W/4
H/8,W/8
H/16,W/16
H/32,W/32
X1
X2
X4
EDB_1
EDB_2
EEM_2
EEM_1
X2_Refined
X1_Refined
Semantic Path
Edge Path
EDB and EEM Modules
Tsem
Semantic Tokenizer
Token sets
Concat
Tedge
Edge Tokenizer
Transformer Encoder
Key/Value
Tsem New
Split
Context_rich Tokens
Tegde New
Main Decoder
Edge Decoder
Refined Semantic Features
EEM
EDB
Concat
3 ×3 Conv
Prediction Heads
Labels
Canny
Edge Labels
Edge Results
Ground truth
Joint Supervision
Fin
Conv
BatchNorm
ReLU
Ffeat
Fout
Aedge
Conv feat
Fedge
1×1 Conv
$
GlobalAvgPool
MLP
AC
AS
Conv1×1
Upsample
Sigmoid
Spatial Attention
Channel Attention
Edge Detection Block
Enhanced Edge Module